

Biotechnology and Food Engineering

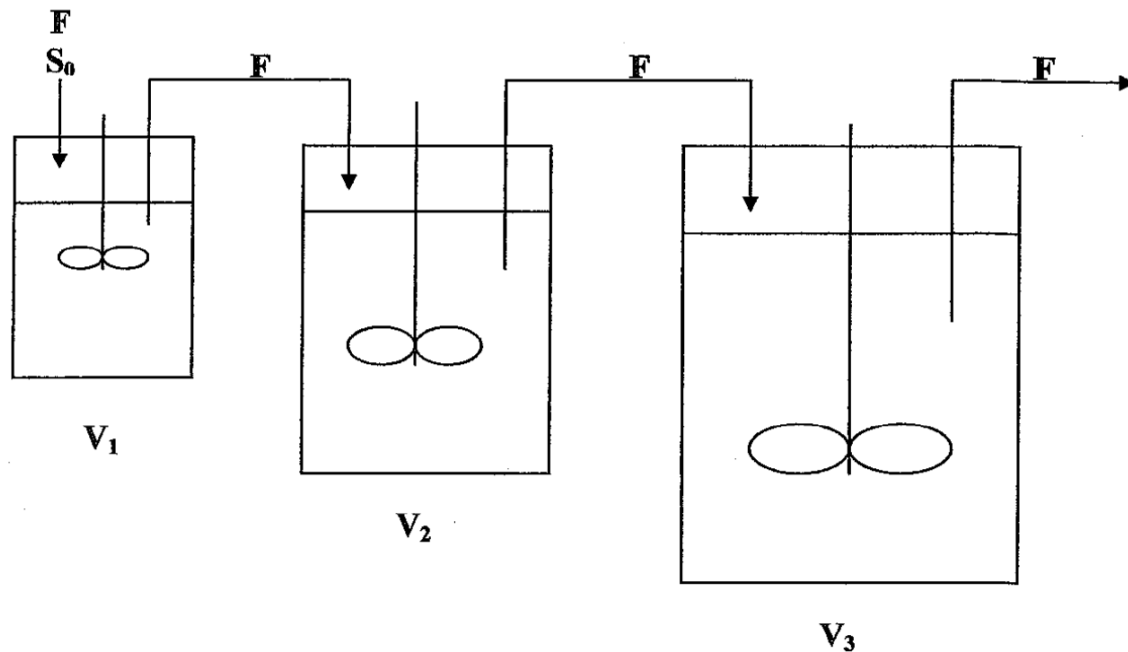
Process Biotechnology

Spring 2026

Tutorial 9 - multi-stage fermentation systems - Solutions

Question 1

A factory performs continuous fermentation to produce a certain product, in a series of 3 fermenters with increasing volumes, that connected serially one after the other, as described in the scheme below:



The fresh feed enters the first fermenter in flow rate of $100 \frac{\text{lit}}{\text{hr}}$, and contains a singular carbon source in concentration of 6%. The biomass yield on this carbon source is $0.47 \frac{\text{gX}}{\text{gS}}$. The volume of the first fermenter is 400 liters, the volume of the second fermenter is 800 liter and the volume of the third and last fermenter is 1 m^3 . In the stream which exits the last fermenter, there is 3.36 ppm leftovers of substrate, and it transferred continuously to separation processes that aim to isolate the product. The separation process yield is 90.91%,

the maximal growth rate of the microorganism in the system is 0.3 per hour and Monod constant equals to $500 \frac{mg}{lit}$. The product yield by the microorganism is $0.39 \frac{gP}{gX}$. It can be assumed that the death rate is negligible, as well as the substrate consumption for maintenance and the product decomposition, and that the product is mostly nitrogen-based.

- A. Develop equations to present the growth rate, biomass concentration and the product concentration in the third fermenter.
- B. What is the growth rate in the third fermenter?
- C. What is the biomass concentration in the second fermenter?
- D. What is the flow rate of the clean protein ($\frac{kg}{hr}$) leaving the separation processes?

Solution:

Given:

$$F = 100 \frac{lit}{hr}$$

$$S_0 = 60 \frac{gS}{lit}$$

$$Y_{\frac{X}{S}} = 0.47 \frac{gX}{gS}$$

$$V_1 = 400 \text{ lit}, V_2 = 800 \text{ lit}, V_3 = 1000 \text{ lit}$$

$$S_{F_3} = 0.0036 \frac{gS}{lit}$$

$$\eta = 90.91\% = 0.9091$$

$$\mu_{max} = 0.3 \text{ hr}^{-1}$$

$$K_s = 0.5 \frac{gS}{lit}$$

$$Y_{\frac{P}{X}} = 0.39 \frac{gP}{gX}$$

$$\mathbf{A. \mu_3, X_{F_3}, P_{F_3} = ?}$$

Fermenter (1):

Standard continuous fermentation, therefore:

$$\mu_1 = D_1$$

$$X_{F_1} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_1})$$

$$P_{F_1} = Y_{\frac{P}{X}} \cdot X_{F_1} = Y_{\frac{P}{S}} \cdot (S_0 - S_{F_1})$$

Fermenter (2):

$$\frac{dX_2}{dt} = \frac{F}{V_2} \cdot X_{F_1} + \mu_2 \cdot X_{F_2} - \cancel{\alpha \cdot X_{F_2}} - \frac{F}{V_2} \cdot X_{F_2} = 0$$

$$D_2 \cdot X_{F_1} + \mu_2 \cdot X_{F_2} - D_2 \cdot X_{F_2} = 0$$

$$\mu_2 \cdot X_{F_2} = D_2 \cdot (X_{F_2} - X_{F_1})$$

$$\mu_2 = D_2 \cdot \left(1 - \frac{X_{F_1}}{X_{F_2}}\right)$$

$$\frac{dS_2}{dt} = \frac{F}{V_2} \cdot S_{F_1} + \frac{\mu_2 \cdot X_{F_2}}{Y_{\frac{X}{S}}} - \frac{\cancel{q_p \cdot X_{F_2}}}{\cancel{\frac{Y_P}{S}}} - \cancel{m \cdot X} - \frac{F}{V_2} \cdot S_{F_2} = 0$$

$$D_2 \cdot S_{F_1} - \frac{\mu_2 \cdot X_{F_2}}{Y_{\frac{X}{S}}} - D_2 \cdot S_{F_2} = 0$$

$$\frac{\mu_2 \cdot X_{F_2}}{Y_{\frac{X}{S}}} = D_2 \cdot (S_{F_1} - S_{F_2})$$

Place the expression for μ_2 that we found above and reduce the equation to:

$$\frac{X_{F_2} - X_{F_1}}{Y_{\frac{X}{S}}} = (S_{F_1} - S_{F_2})$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (S_{F_1} - S_{F_2}) + X_{F_1}$$

Place X_{F_1} that was found above for fermenter (1):

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (\cancel{S_{F_1}} - S_{F_2}) + Y_{\frac{X}{S}} \cdot (S_0 - \cancel{S_{F_1}})$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_2})$$

$$\frac{dP_2}{dt} = \frac{F}{V_2} \cdot P_{F_1} + q_p \cdot X_{F_2} - \cancel{K \cdot P_{F_2}} - \frac{F}{V_2} \cdot P_{F_2} = 0$$

$$D_2 \cdot P_{F_1} - q_p \cdot X_{F_2} - D_2 \cdot P_{F_2} = 0$$

$$\text{Place } q_p = \mu_2 \cdot Y_{\frac{P}{X}}:$$

$$\mu_2 \cdot Y_{\frac{P}{X}} \cdot X_{F_2} = D_2 \cdot (P_{F_2} - P_{F_1})$$

$$\text{Place } \mu_2 \cdot X_{F_2}:$$

$$Y_{\frac{P}{X}} \cdot \cancel{D_2} \cdot (X_{F_2} - X_{F_1}) = \cancel{D_2} \cdot (P_{F_2} - P_{F_1})$$

$$(X_{F_2} - X_{F_1}) \cdot Y_{\frac{P}{X}} = P_{F_2} - Y_{\frac{P}{X}} \cdot X_{F_1}$$

$$P_{F_2} = Y_{\frac{P}{X}} \cdot X_{F_2} - \cancel{Y_{\frac{P}{X}} \cdot X_{F_1}} + \cancel{Y_{\frac{P}{X}} \cdot X_{F_1}} \rightarrow P_{F_2} = Y_{\frac{P}{X}} \cdot X_{F_2}$$

$$\text{Place } X_{F_2}:$$

$$P_{F_2} = Y_{\frac{P}{S}} \cdot (S_0 - S_{F_2})$$

Fermenter (3):

In a similar way:

$$\frac{dX_3}{dt} = \frac{F}{V_3} \cdot X_{F_2} + \mu_3 \cdot X_{F_3} - \frac{F}{V_3} \cdot X_{F_3} = 0$$

$$\mu_3 = D_3 \cdot \left(1 - \frac{X_{F_2}}{X_{F_3}}\right)$$

$$\frac{dS_3}{dt} = \frac{F}{V_3} \cdot S_{F_2} + \frac{\mu_3 \cdot X_{F_3}}{Y_{X/S}} - \frac{F}{V_3} \cdot S_{F_3}$$

$$X_{F_3} = Y_{X/S} \cdot (S_0 - S_{F_3})$$

$$\frac{dP_3}{dt} = \frac{F}{V_3} \cdot P_{F_2} + q_p \cdot X_{F_3} - \frac{F}{V_3} \cdot P_{F_3} = 0$$

$$P_{F_3} = Y_{P/S} \cdot (S_0 - S_{F_3})$$

B. $\mu = ?$

We found that: $\mu_3 = D_3 \cdot \left(1 - \frac{X_{F_2}}{X_{F_3}}\right)$, but we can't calculate it, so we will turn to Monod equation:

$$\mu = \mu_{max} \left[\frac{S_{F_3}}{K_s + S_{F_3}} \right]$$

$$\mu_3 = 0.3 \text{ hr}^{-1} \left[\frac{0.0036 \frac{gS}{lit}}{0.5 + 0.0036 \frac{gS}{lit}} \right]$$

$$\mu_3 = 0.002 \text{ hr}^{-1}$$

C. $X_{F_2} = ?$

We previously found that: $X_{F_2} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_2})$ but we don't have S_{F_2} , although we do

have S_{F_3} . Therefore we can get to X_{F_2} from $\mu_3 = D_3 \cdot (1 - \frac{X_{F_2}}{X_{F_3}})$:

$$X_{F_3} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_3}) = 0.47 \frac{gX}{gS} \cdot (60 - 0.0036) \frac{gS}{lit} = 28.2 \frac{gX}{lit}$$

$$D_3 = \frac{F}{V_3} = \frac{100 \frac{lit}{hr}}{1000 lit} = 0.1 \frac{1}{hr}$$

Place all numbers to find X_{F_2} :

$$0.002 \frac{1}{hr} = 0.1 \frac{1}{hr} \cdot \left(1 - \frac{X_{F_2}}{28.2 \frac{gX}{lit}} \right)$$

$$X_{F_2} = 27.64 \frac{gX}{lit}$$

D. $G_{P_{clean}} = ?$

$$G_{P_{clean}} = F \cdot P_{F_3} \cdot \eta$$

$$P_{F_3} = Y_{\frac{P}{X}} \cdot X_{F_3} = 0.39 \frac{gP}{gX} \cdot 28.2 \frac{gX}{lit} = 11 \frac{gP}{lit}$$

$$G_{P_{clean}} = 100 \frac{lit}{hr} \cdot 11 \frac{gP}{lit} \cdot 0.9091 = 1000 \frac{gP}{hr}$$

$$G_{P_{clean}} = 1 \frac{kgP}{hr}$$

Question 2

A factory performs continuous multi-phase fermentation for the production of a certain extracellular product. The fermentation includes three phases connected serially to one another:

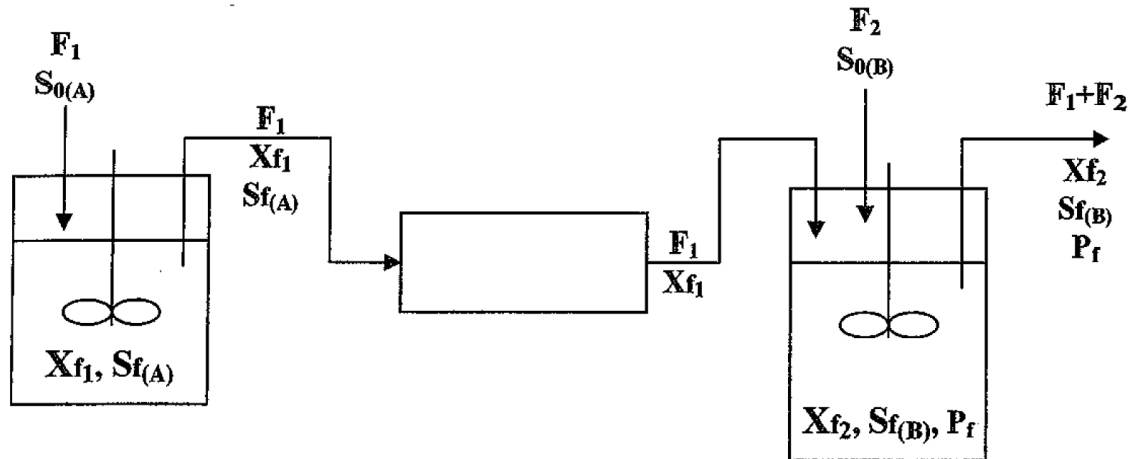
First phase: bacteria growth in standard continuous fermenter (CSTR), without product production, only biomass growth. The fermenter volume is 75 m^3 and it is fed constantly with solution of substrate A at a concentration of 1.75%. The growth yield on substrate A is $0.594 \frac{\text{gX}}{\text{gS}}$, the maximal growth rate is 0.2 hr^{-1} and Monod constant on substrate A is 450 ppm.

Second phase: the liquid exiting the first fermenter moves into a continuous plug flow fermenter (CPFR). Since the bacteria growth is done only in aerobic conditions - there is no growth in this fermenter, but all the remaining of substrate A in the medium is completely consumed for the maintenance of the bacteria. The volume of this fermenter is 1.5 m^3 and each bacterium stay in this fermenter for 10 minutes (from the moment it enters to the fermenter until it exits).

Third phase: the liquid leaves the plug flow fermenter enters into another standard continuous fermenter (CSTR), with the volume of 20 m^3 . This fermenter is also fed with fresh feed at a flow rate of $500 \frac{\text{lit}}{\text{hr}}$, that contains substrate B in concentration of 10%. Substrate B is mostly used for product production ($Y_{\frac{P}{S_B}} = 0.24 \frac{\text{gP}}{\text{gS}}$), but also used for the growth of the bacteria. Substrate B concentration in the flow exiting this fermenter is $2 \frac{\text{g}}{\text{lit}}$ and the productivity of the product production in this fermenter is $0.285 \frac{\text{gP}}{\text{lit} \cdot \text{hr}}$.

- A. What is the mass flow rate ($\frac{\text{kg}}{\text{hr}}$) of the product that exits the third fermenter?**
- B. In another fermentation, this bacterium was grown on a medium with substrate B as the carbon source, and the substrate was consumed for biomass growth only without producing any product. In this fermentation it was found that the substrate yield on the biomass is $0.725 \frac{\text{gS}}{\text{gX}}$.**

What is the mass flow rate ($\frac{\text{kg}}{\text{hr}}$) of the total biomass that exit the third fermenter?



Solution:

Given:

<u>CSTR 1</u>	<u>CPFR 2</u>	<u>CSTR 3</u>
$V_1 = 75,000 \text{ lit}$	$S_{F_2(A)} = 17.5 \frac{gS}{lit}$	$V_3 = 20,000 \text{ lit}$
$S_{0(A)} = 17.5 \frac{gS}{lit}$	$V_2 = 1,500 \text{ lit}$	$F_2 = 500 \frac{lit}{hr}$
$K_s = 0.45 \frac{g}{lit}$	$t = 10 \text{ min} = 0.167 \text{ hr}$	$S_{0(B)} = 100 \frac{gS}{lit}$
$Y_{\frac{X}{S(A)}} = 0.594 \frac{gX}{gS}$		$Y_{\frac{P}{S(B)}} = 0.24 \frac{gP}{gS}$
$\mu_{max} = 0.2 \frac{1}{hr}$		$S_{F_3(B)} = 2 \frac{gS}{lit}$
		$P_r = 0.285 \frac{gP}{lit \cdot hr}$

A. $G_{P_3} = ?$

$$G_{P_3} = (F_1 + F_2) \cdot P_{F_3}$$

F_2 is given in the question, F_1 can be easily calculated from the parameters given in the plunger fermenter, as follow:

$$t_{staying} = \frac{V_2}{F_1} \rightarrow F_1 = \frac{1,500 \text{ lit}}{0.167 \text{ hr}} = 9,000 \frac{lit}{hr}$$

We have left to calculate P_{F_3} , that can be achieved by using the product balance:

$$\frac{dP_3}{dt} = \frac{F_1}{V_3} \cdot P_{F_1} + \frac{F_2}{V_3} \cdot P_{F_2} + q_P \cdot X_{F_3} - \frac{F_1 + F_2}{V_3} \cdot P_{F_3} = 0$$

$$q_P \cdot X_{F_3} = \left(\frac{F_1 + F_2}{V_3} \right) \cdot P_{F_3}$$

The expression $\left(\frac{F_1 + F_2}{V_3} \right) \cdot P_{F_3}$ is basically the productivity (P_r) which is given in the question:

$$\left(\frac{F_1 + F_2}{V_3} \right) \cdot P_{F_3} = P_r = 0.285 \frac{gP}{lit \cdot hr}$$

$$P_{F_3} = \frac{P_r}{F_1 + F_2} \cdot V_3 = \frac{0.285 \frac{gP}{lit \cdot hr}}{9,000 \frac{lit}{hr} + 500 \frac{lit}{hr}} \cdot 20,000 lit = 0.6 \frac{gP}{lit}$$

$$G_{P_3} = \left(9,000 \frac{lit}{hr} + 500 \frac{lit}{hr} \right) 0.6 \frac{gP}{lit}$$

$$G_{P_3} = 5,700 \frac{gP}{hr} = 5.7 \frac{kgP}{hr}$$

B. $G_{X_{tot.}}$ = ?

Given for growth purposes fermentation: $Y_{\frac{S}{X}(B)} = 0.725 \frac{gS}{gX}$

$$G_{X_{tot.}} = (F_1 + F_2) \cdot X_{F_2}$$

We need to find X_{F_2} but we can't do it by using the standard balance since we don't have $Y_{\frac{X}{S}}$. The given $Y_{\frac{S}{X}(B)}$ is relevant for fermentation for biomass growth only, while in our case the fermentation has product production, so we can't use the given yield.

From balance on mass flow rates:

$$G_{X_{tot.}} = G_{X_1} + G_{X_3}$$

We need to calculate G_{X_1} and G_{X_3} .

$$G_{X_1} = F_1 \cdot X_{F_1}$$

$$X_{F_1} = Y_X \cdot (S_0 - S_F)$$

$$S_F = \frac{\mu \cdot K_s}{\mu_{max} - \mu}$$

We know K_s and μ_{max} , and we can calculate μ :

$$\mu = \frac{F}{V} = \frac{9,000 \frac{lit}{hr}}{75,000 lit} = 0.12 \frac{1}{hr}$$

$$S_F = \frac{0.12 \frac{1}{hr} \cdot 0.45 \frac{gS}{lit}}{0.2 \frac{1}{hr} - 0.12 \frac{1}{hr}} = 0.675 \frac{gS}{lit}$$

$$X_{F_1} = 0.594 \frac{gX}{gS} \cdot (17.5 - 0.675) \frac{gS}{lit} = 10 \frac{gX}{lit}$$

$$G_{X_1} = 9,000 \frac{lit}{hr} \cdot 10 \frac{gX}{lit} = 90,000 \frac{gX}{hr} = \mathbf{90 \frac{kgX}{hr}}$$

Now we will calculate G_{X_3} :

We have information about the substrate yield on the biomass during fermentation for growth purpose only, $Y_{S(X(B))}$. We want to use it in order to find the mass flow rate in fermenter 3. The relationship between these parameters is the substrate flow rate for biomass only (G_s):

$$G_{X_3} = \frac{G_s(biomass)}{Y_{S(X(B))}}$$

In order to find $G_s(biomass)$ we will perform balance on substrate B flow rates in the third fermenter:

$$G_s(biomass) = G_{S_0} - G_s(product) - G_{S_F}$$

Let's calculate each of the equation components:

$$G_{S_0} = F_2 \cdot S_{0(B)} = 500 \frac{lit}{hr} \cdot 100 \frac{gS}{lit} = \mathbf{50,000 \frac{gS}{hr}}$$

$$G_S(\text{product}) = \frac{G_{P_{F_3}}}{Y_{P/S}}$$

Explanation – the amount of substrate consumed per hour, equals to the amount of product produced in 1 hour divided by the product yield. From an equations point of view:

$$P_F = Y_{\frac{P}{S}} \cdot (S_0 - S_F)$$

As $G_P = P_F \cdot F$, we will multiply this equation by F to get:

$$G_P = Y_{\frac{P}{S}} \cdot (S_0 - S_F) \cdot F$$

$$\frac{G_P}{Y_{\frac{P}{S}}} = F \cdot (S_0 - S_F) = G_S$$

Back to the equation to calculate ($G_{P_{F_3}}$ was calculated above and $Y_{P/S}$ is given in the question):

$$G_S(\text{product}) = \frac{5.7 \frac{kgP}{hr}}{0.24 \frac{gP}{gS}} = 23.75 \frac{kgS}{hr} = \mathbf{23,750 \frac{gS}{hr}}$$

$$G_{S_F} = (F_1 + F_2) \cdot S_{F_3(B)} = 9,500 \frac{lit}{hr} \cdot 2 \frac{gS}{lit} = \mathbf{19,000 \frac{gS}{hr}}$$

$$G_S(\text{biomass}) = (50,000 - 23,750 - 19,000) \frac{gS}{hr} = 7250 \frac{gS}{hr}$$

$$G_{X_3} = \frac{7250 \frac{gS}{hr}}{0.725 \frac{gS}{gX}} = 10,000 \frac{gX}{hr} = \mathbf{10 \frac{kgX}{hr}}$$

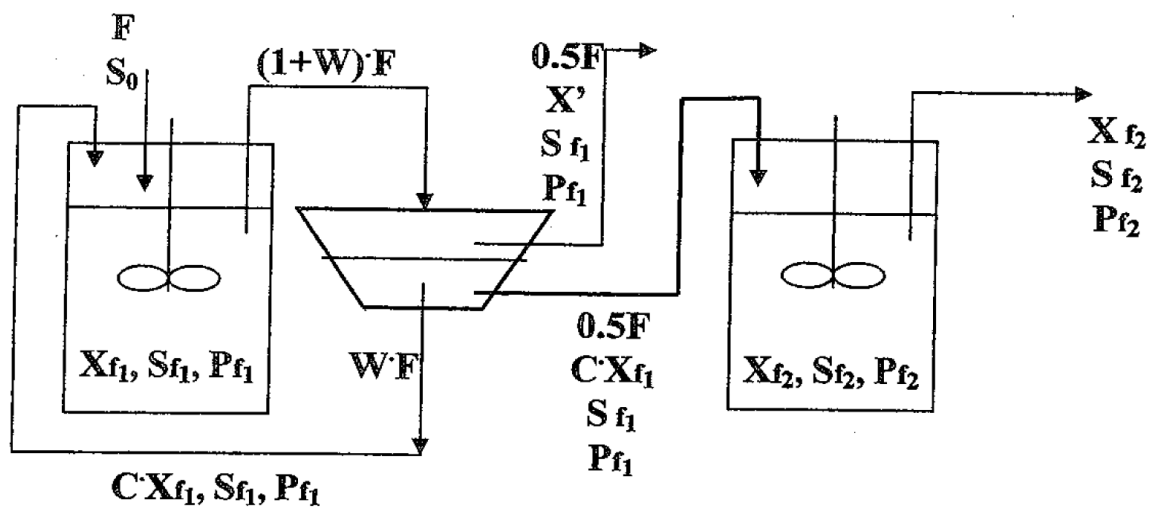
G_{X_3} is the biomass produced in fermenter 3, but we have to remember that this fermenter received in addition all the biomass grown on fermenter 1 (G_{X_1}). Therefore:

$$G_{tot} = G_{X_1} + G_{X_3} = 90 \frac{kgX}{hr} + 10 \frac{kgX}{hr}$$

$$G_{tot} = \mathbf{100 \frac{kgX}{hr}}$$

Question 3

A system of 2 continuous fermenters with the same volume, is connected serially as described hereunder. The first is fermenter with circulation and concentration, while the second is standard continuous fermenter. The first fermenter receives external fresh feed, and the broth of this fermenter is streaming to a continuous separator. The top flow exits the separator at a flow rate of $0.5 \cdot F$, and continues to separation processes of the product. From the bottom of the separator exiting two flows: one at a flow rate of $0.5 \cdot F$ entering into the second fermenter and constitute its “fresh” feed, and the other flows at a rate of $W \cdot F$ which is the circulation flow returning to the first fermenter. Below is a scheme of the system:



It is possible to use all the standard assumption that were used during the equation development, except for assumptions that is being contradicted in the system described above.

- A. Develop the biomass balance ($\frac{dX}{dt}$) until receiving an equation describes the growth rate in steady state in the second fermenter. Continue to develop the equation until reaching the simplest equation, with minimal unknown parameters.
- B. Develop the substrate balance ($\frac{dS}{dt}$) until receiving an equation describes the biomass concentration in steady state in the second fermenter. Continue to develop the equation until reaching the simplest equation, with minimal unknown parameters.

- C. Develop the product balance ($\frac{dP}{dt}$) until receiving an equation describes the product concentration in steady state in the second fermenter. Continue to develop the equation until reaching the simplest equation, with minimal unknown parameters.
- D. If the first fermenter was continuous fermenter with circulation and without concentration (*i.e.* $c = 1$), how will the equation developed above be updated? Write the updated equations and explain why and how they are similar to the other equations which we developed in class.

Solution:

A. $\mu_2 = ?$

Fermenter (1):

Continuous fermentation with circulation and concentration, therefore:

$$\mu_1 = D \cdot [1 + w(1 - C)]$$

Fermenter (2):

$$\frac{dX_2}{dt} = \frac{0.5 \cdot F}{V} \cdot C \cdot X_{F_1} + \mu_2 \cdot X_{F_2} - \cancel{\alpha_2 \cdot X_{F_2}} - \frac{0.5 \cdot F}{V} \cdot X_{F_2} = 0$$

$$\mu_2 \cdot X_{F_2} = 0.5 \cdot D \cdot X_{F_2} - 0.5 \cdot D \cdot C \cdot X_{F_1} = 0.5 \cdot D \cdot (X_{F_2} - C \cdot X_{F_1})$$

$$\mu_2 = 0.5 \cdot D \cdot \left(1 - \frac{C \cdot X_{F_1}}{X_{F_2}} \right)$$

B. $X_{F_2} = ?$

Fermenter (1):

Continuous fermentation with circulation and concentration, therefore:

$$X_{F_1} = \frac{Y_X \cdot \left(S_0 - S_{F_1} \right)}{\frac{\bar{S}}{[1 + w \cdot (1 - C)]}}$$

Fermenter (2):

$$\frac{dS_2}{dt} = \frac{0.5 \cdot F}{V} \cdot S_{F_1} - \frac{\mu_2 \cdot X_{F_2}}{Y_{\frac{X}{S}}} - \frac{\cancel{q_P \cdot X_{F_2}}}{\cancel{Y_P}} - \cancel{m_2 \cdot X_{F_2}} - \frac{0.5 \cdot F}{V} \cdot S_{F_2} = 0$$

$$\frac{\mu_2 \cdot X_{F_2}}{Y_{\frac{X}{S}}} = 0.5 \cdot D \cdot (S_{F_1} - S_{F_2})$$

Place the expression we found on section A for $\mu_2 \cdot X_{F_2}$:

$$\frac{0.5 \cdot D \cdot (X_{F_2} - C \cdot X_{F_1})}{Y_{\frac{X}{S}}} = 0.5 \cdot D \cdot (S_{F_1} - S_{F_2})$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (S_{F_1} - S_{F_2}) + C \cdot X_{F_1}$$

Place X_{F_1} that was found above:

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (S_{F_1} - S_{F_2}) + C \cdot \left(\frac{Y_{\frac{X}{S}} \cdot (S_0 - S_{F_1})}{[1 + w \cdot (1 - C)]} \right)$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot \left[S_{F_1} - S_{F_2} + C \cdot \left(\frac{(S_0 - S_{F_1})}{[1 + w \cdot (1 - C)]} \right) \right]$$

C. $P_{F_2} = ?$

Fermenter (1):

$$P_{F_1} = Y_{\frac{P}{X}} \cdot [1 + w \cdot (1 - c)] \cdot X_{F_1} = Y_{\frac{P}{S}} \cdot (S_0 - S_{F_1})$$

Fermenter (2):

$$\frac{dP_2}{dt} = \frac{0.5 \cdot F}{V} \cdot P_{F_1} - q_{P_2} \cdot X_{F_2} - \frac{\cancel{K \cdot P_{F_2}}}{\cancel{Y_P}} - \frac{0.5 \cdot F}{V} \cdot P_{F_2} = 0$$

$$\mu_2 \cdot Y_{\frac{P}{X}} \cdot X_{F_2} = 0.5 \cdot D \cdot (P_{F_2} - P_{F_1})$$

Place $\mu_2 X_{F_2}$ that was found previously:

$$0.5 \cdot D \cdot (X_{F_2} - C \cdot X_{F_1}) \cdot Y_{\frac{P}{X}} = 0.5 \cdot D \cdot (P_{F_2} - P_{F_1})$$

$$P_{F_2} = Y_{\frac{P}{X}} \cdot (X_{F_2} - C \cdot X_{F_1}) + P_{F_1}$$

Place P_{F_1} as was found above:

$$P_{F_2} = Y_{\frac{P}{X}} \cdot (X_{F_2} - C \cdot X_{F_1}) + Y_{\frac{P}{X}} \cdot [1 + w \cdot (1 - c)] \cdot X_{F_1}$$

$$P_{F_2} = Y_{\frac{P}{X}} \cdot \{X_{F_2} + X_{F_1} \cdot [1 + w \cdot (1 - C) - C]\}$$

D.

Fermenter (1):

$$\mu_1 = D_1 \cdot (1 + w(1 - C))$$

It is given that $C = 1$, therefore: $\mu_1 = D_1$

$$X_{F_1} = \frac{Y_X(S_0 - S_{F_1})}{1 + w(1 - C)}$$

$$X_{F_1} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_1})$$

Fermenter (2):

$$\mu_2 = 0.5 \cdot D \cdot \left(1 - \frac{C X_{F_1}}{X_{F_2}}\right)$$

$$\mu_2 = 0.5 \cdot D \cdot \left(1 - \frac{X_{F_1}}{X_{F_2}}\right)$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot \left[S_{F_1} - S_{F_2} + C \cdot \left(\frac{(S_0 - S_{F_1})}{1 + w(1 - C)} \right) \right]$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot [S_0 - S_{F_2}]$$

$$P_{F_2} = Y_{\frac{P}{X}} \cdot \{X_{F_2} + X_{F_1} [1 + w \cdot (1 - C) - C]\}$$

$$P_{F_2} = Y_{\frac{P}{X}} \cdot X_{F_2}$$